

Cross-Platform Colon Cancer Proliferation Classification

Leakage-Free ML Pipeline | StabilitySelector | Calibration Benchmark | Drug Screen

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Problem & Motivation

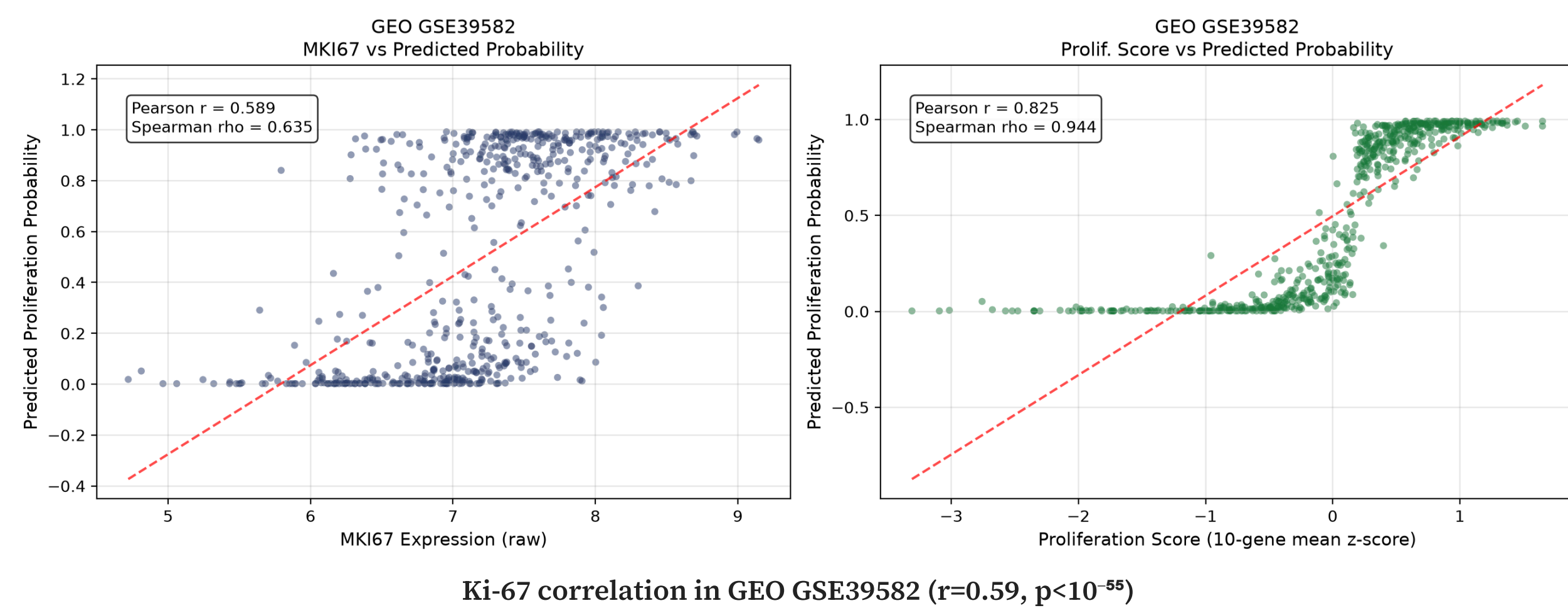
Colon cancer is the 2nd leading cause of cancer death worldwide. Ki-67 IHC — the clinical standard for proliferation assessment — is subjective, provides only a single-gene readout, and offers no drug prediction.

Can ML predict proliferation from transcriptomics without data leakage?

We address this with a leakage-free pipeline trained on GEO (Affymetrix, n=585+232), validated on TCGA-COAD (RNA-seq, n=329) and CPTAC (proteomics, n=105), and linked to drug sensitivity via GDSC2 (295 drugs × 969 cell lines).

Key Innovation: A bootstrap-based StabilitySelector that retains features present in top-K across ≥50% of 100 resamples, combined with nested cross-validation and anti-leakage gene filtering.

Ki-67 Correlation (Leakage-Free)

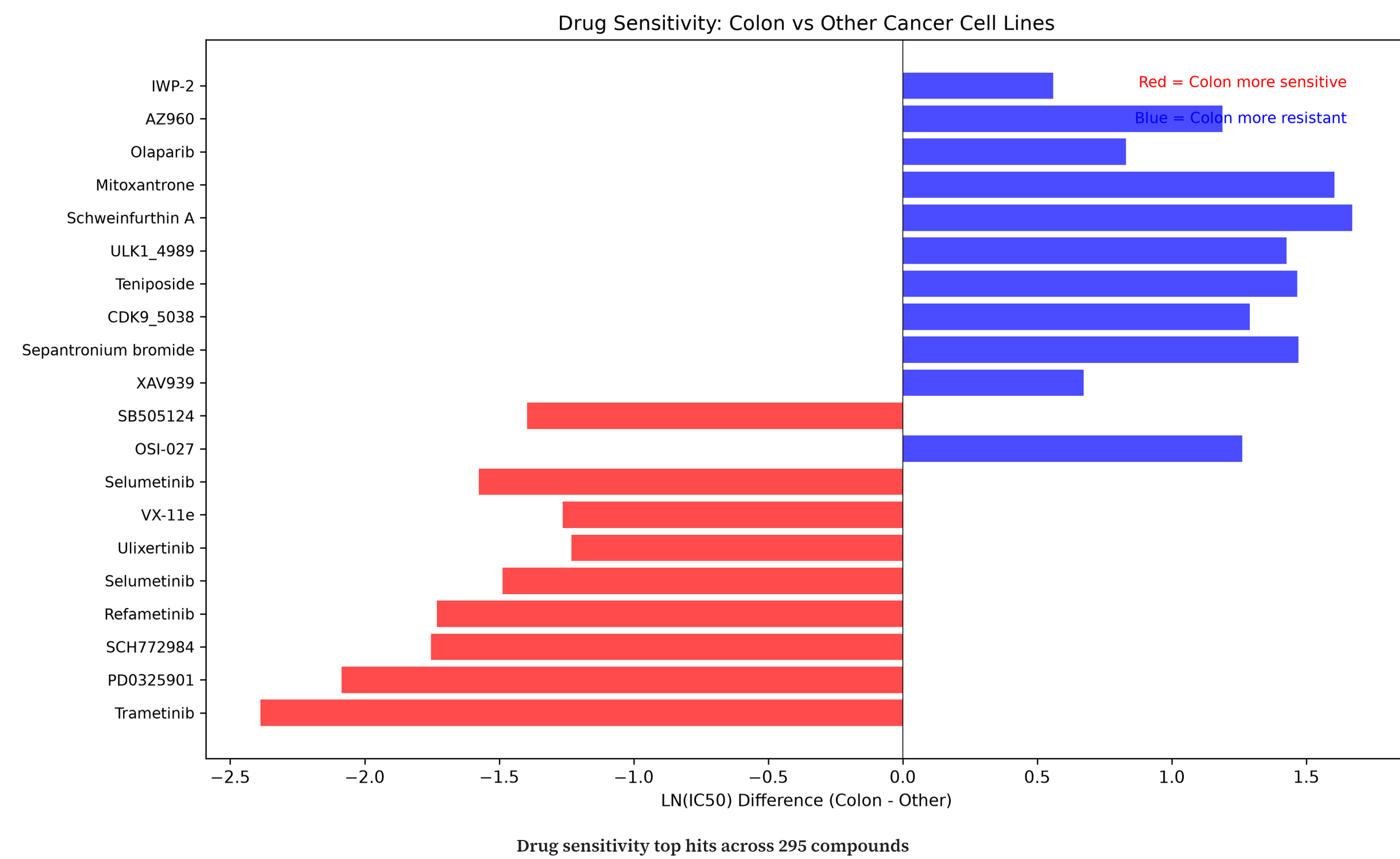


Ki-67 correlation in GEO GSE39582 ($r=0.59$, $p<10^{-65}$)

- GEO:** $r = 0.59$, $p < 10^{-65}$
- TCGA-COAD:** $r = 0.54$, $p < 10^{-25}$

10 proliferation genes (MKI67, TOP2A, etc.) removed from feature set. Model infers proliferation via downstream transcriptional cascades — not direct Ki-67 proxies.

Drug Sensitivity (GDSC2)



Bonferroni-corrected: 295 drugs × 969 cell lines

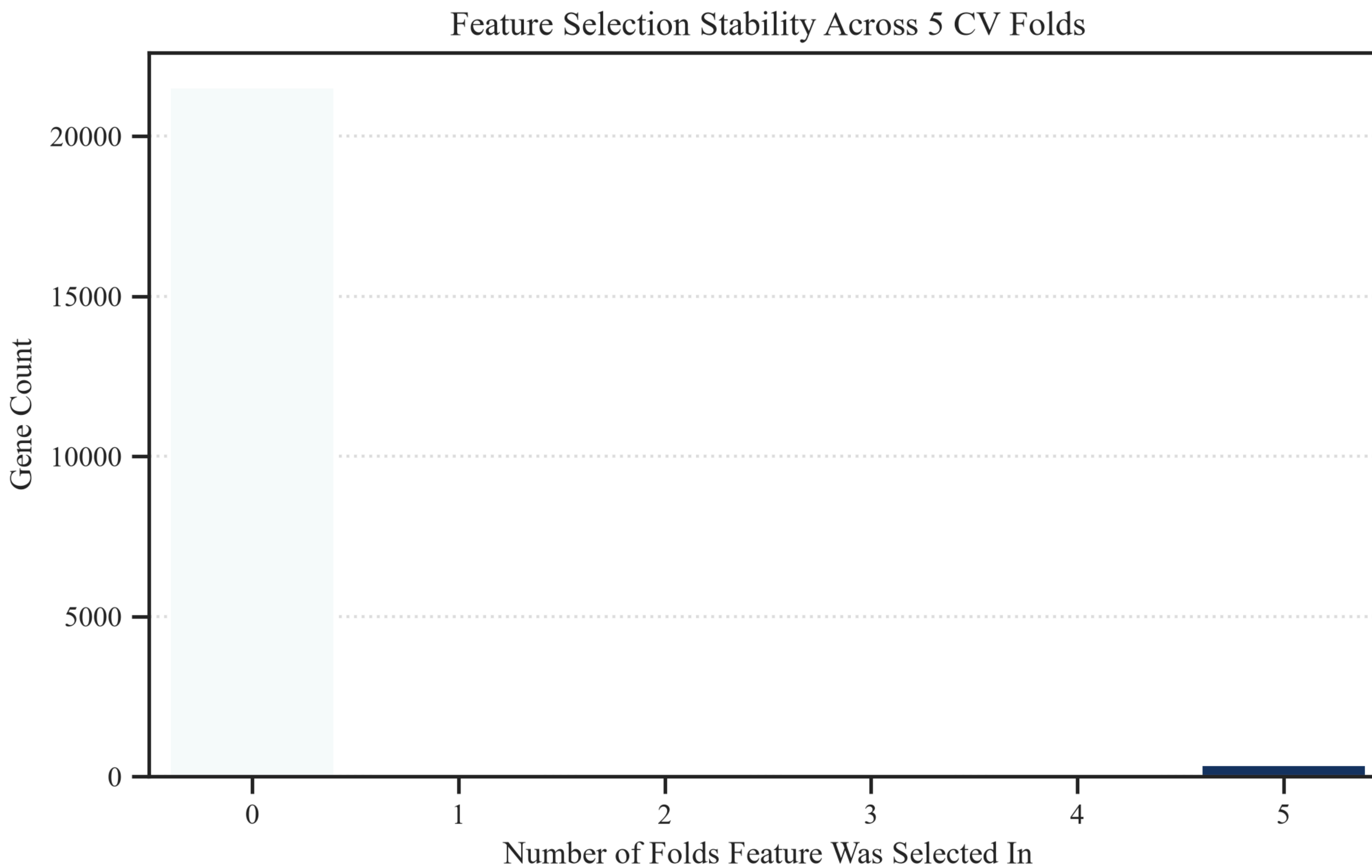
- Trametinib (MEK)** — $p = 1.8 \times 10^{-12}$
- PD0325901 (MEK)** — $p = 5.9 \times 10^{-12}$
- SCH772984 (ERK)** — $p = 1.1 \times 10^{-10}$
- Refametinib (MEK)** — $p = 2.7 \times 10^{-10}$

Pipeline & Methodology

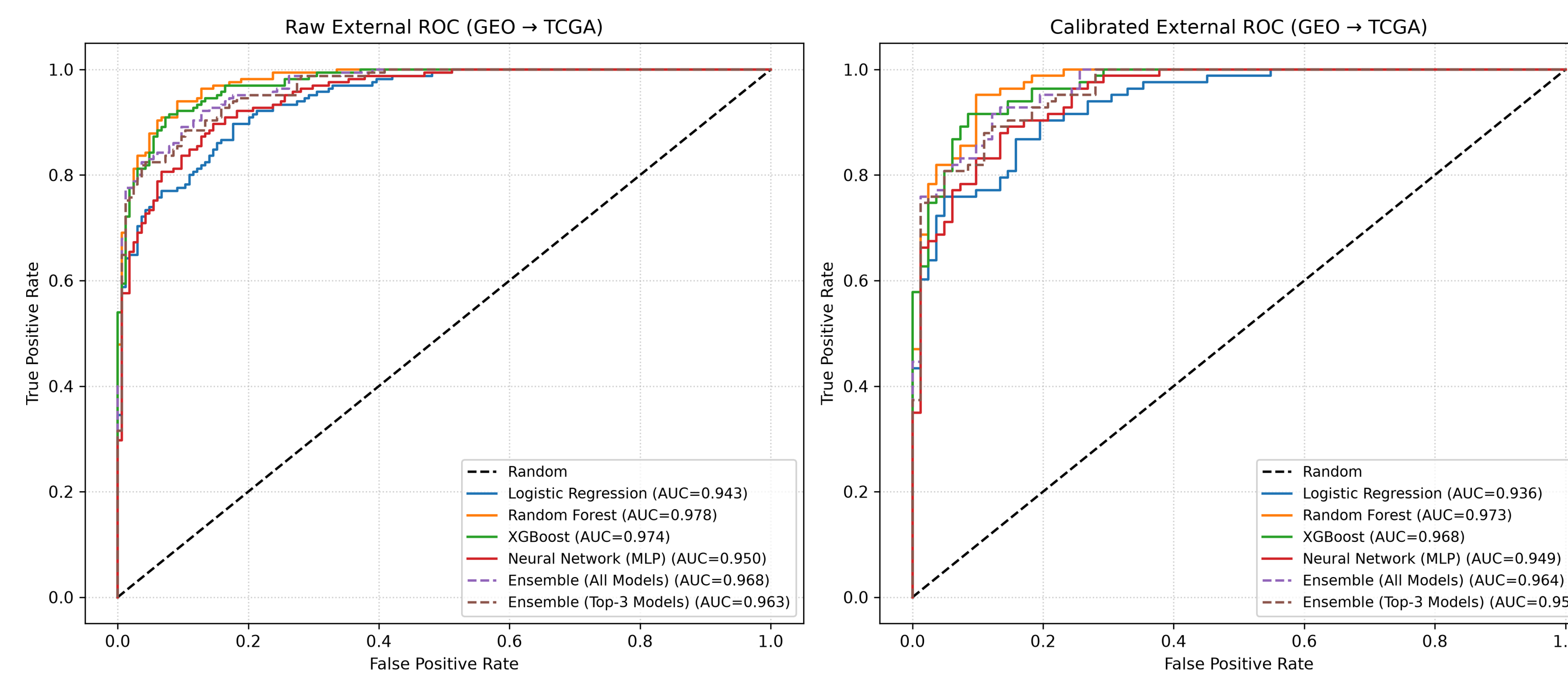
- GEO GSE39582 (585) + GSE17538 (232) → Train split
- Remove 10 proliferation genes (anti-leakage guard)
- StabilitySelector: bootstrap 100×, keep ≥50% consensus
- 4 models: LR, RF, XGBoost, MLP with nested CV
- TCGA-COAD (329) + CPTAC (105) → External validation
- 5 calibration methods (Platt, Isotonic, QN, QN+Platt)
- GDSC2: 295 drugs × 969 lines (Bonferroni corrected)

StabilitySelector: Novel sklearn Transformer

Bootstrap 100× | ANOVA F-scores per resample | Retain features in top-K across ≥50% of resamples



External Validation: GEO → TCGA



External ROC curves from GEO-trained models on TCGA-COAD

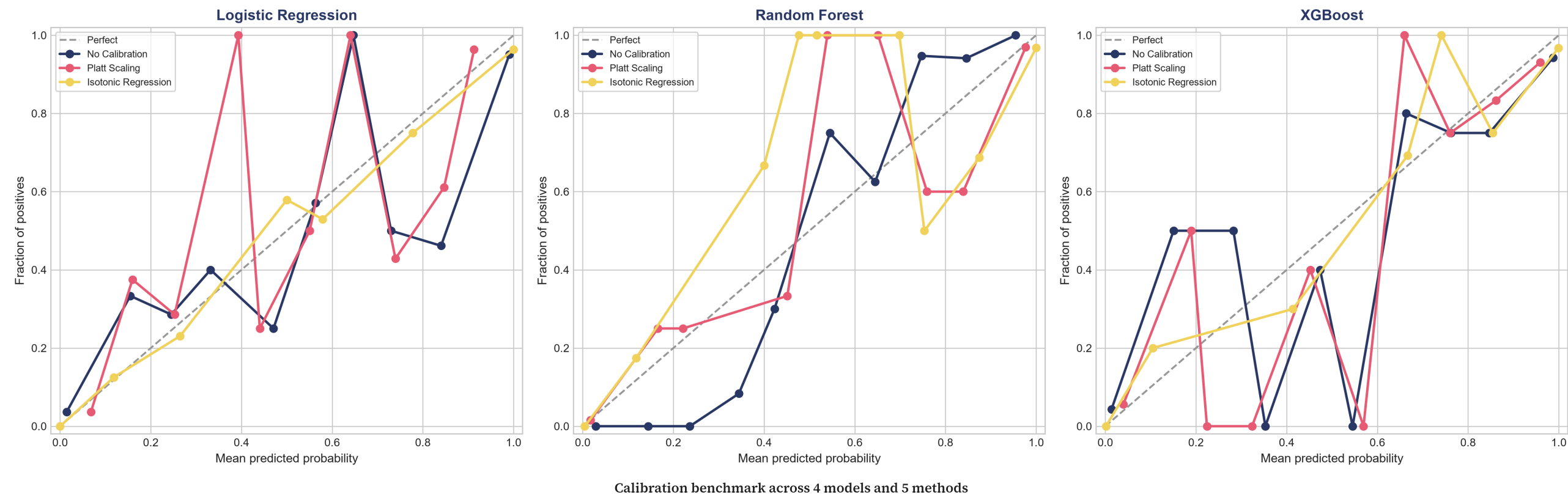
Cross-Platform AUCs (GEO Affymetrix → TCGA RNA-seq)

Model	Raw AUC	Calibrated AUC
Logistic Regression	0.936	0.937
Random Forest	0.973	0.973
XGBoost	0.968	0.968
MLP	0.949	0.949

Calibration Benchmark Results

Key Finding: Platt Scaling Significantly Reduces ECE

- RF Platt: AUC=0.973, ECE=0.043 — 95% CIs non-overlapping with uncalibrated (ECE=0.115)
- LR QN+Platt: AUC=0.972 — best cross-platform transfer calibration



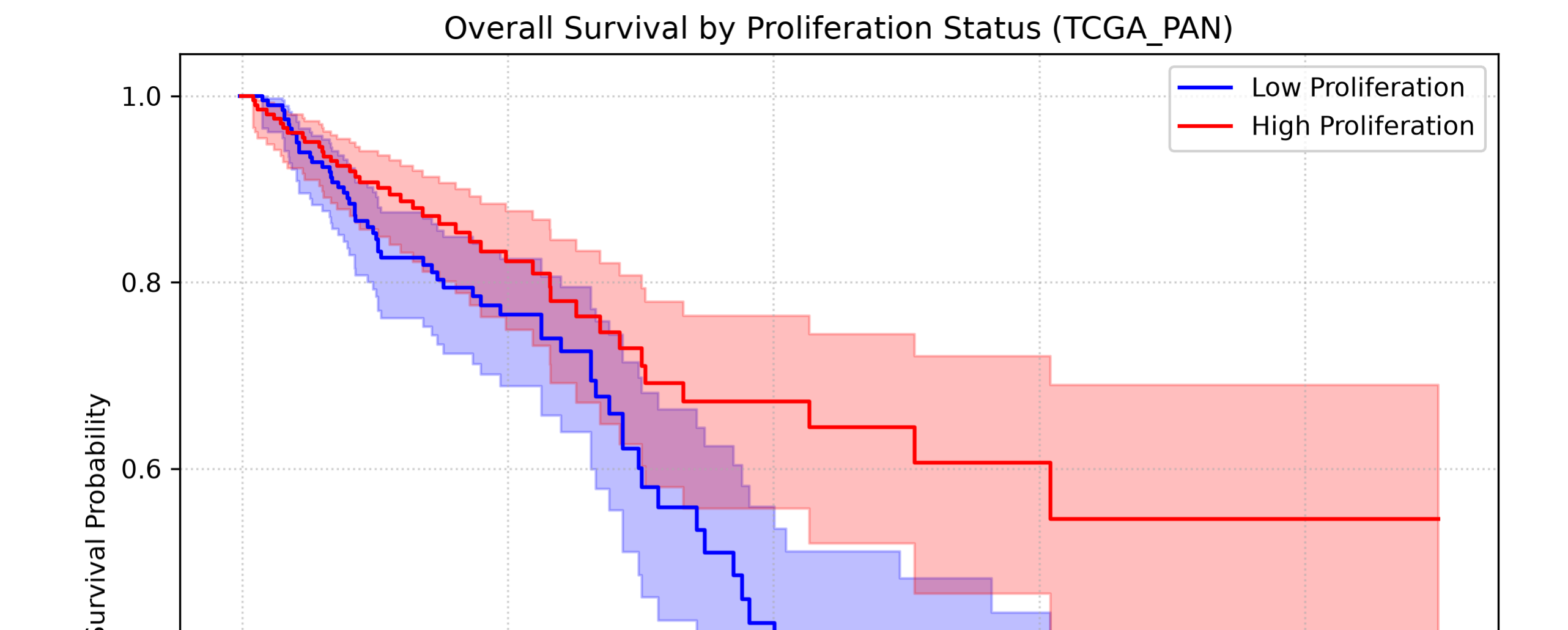
Best Config per Model (sorted by ECE)

Ablation: StabilitySelector vs SelectKBest

Selector	Model	Holdout AUC
SS	Logistic Regression	0.994
SKB	Logistic Regression	0.994
SS	Random Forest	0.988
SKB	Random Forest	0.988
SS	XGBoost	0.990
SKB	XGBoost	0.991
SS	MLP	0.992
SKB	MLP	0.983

SS matches or beats SKB on all 4 models. **MLP: +0.009 AUC (0.983 → 0.992).**

Survival Analysis



Limitations & Future Work

- Cox PH borderline** ($p=0.092$) — larger cohorts needed
- CPTAC underpowered** — only 7 survival events
- Drug sensitivity** uses tissue type as proxy; future work will stratify by predicted proliferation score within colon lines
- Prospective validation** needed before clinical translation
- Multi-cancer extension** — apply 10-gene signature to pan-cancer datasets
- Deep learning** — test transformer models on raw expression (e.g., Geneformer)

References (Selected)

- Marisa et al. *PLoS Med* 2013 — GSE39582
- Smith et al. *Gastroenterology* 2010 — GSE17538
- TCGA Network. *Nature* 2012 — TCGA-COAD
- Whitfield et al. *MBOC* 2002 — Cell cycle genes
- Platt. *NIPS* 1999 — Platt scaling
- Meinshausen & Bühlmann. *JRSS-B* 2010 — Stability selection
- Iorio et al. *Cell* 2016 — GDSC2
- Niculescu-Mizil & Caruana. *ICML* 2005 — Calibration
- Zhang et al. *Nature* 2014 — CPTAC
- Van Calster et al. *BMC Med* 2019 — Clinical calibration
- Full list:** github.com/Ronisnotasianfr/ColoGrowth-ML